AUSTRALIAN MATHS TRUST

AUSTRALIAN MATHEMATICS COMPETITION

Chinese Version • Junior

Australian school years 7 and 8

2019

NAME: _____

TIME ALLOWED: 75 minutes

INSTRUCTIONS AND INFORMATION

General

1 Do not open the booklet until told to do so by your teacher.

2 NO calculators, maths stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.

3 Diagrams are NOT drawn to scale. They are intended only as aids.

4 There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.

5 This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.

6 Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.

7 When your teacher gives the signal, begin working on the problems.

The answer sheet

1 Use only lead pencil.

2 Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.

3 Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder: You may sit this competition once, in one division only, or risk no score.

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1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 算式 $201 - 9 =$

$201 - 9 =$

(A) 111 (B) 182 (C) 188 (D) 192 (E) 198

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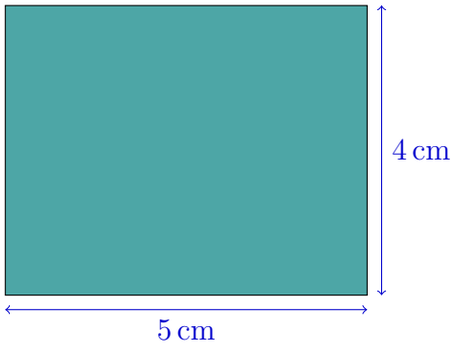
2. 这个矩形的长为 5 cm、宽为 4 cm。
请问它的面积为多少 cm^2？

This rectangle is 5 cm wide and 4 cm tall.
What is its area in square centimetres?

(A) 9 (B) 10 (C) 18 (D) 20 (E) 40



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3. 右表显示 5 年级男生与女生年龄为 10 或 11 岁的人数。
请问 5 年级男生中有多少人为 11 岁？

	10 岁	11 岁	合计
女生	14	25	39
男生	9	?	46
合计	23	62	85

The table shows the number of boys and girls aged 10 or 11 in year 5.
How many boys aged 11 are in year 5?

(A) 9 (B) 11 (C) 21 (D) 37 (E) 46

	Age 10	Age 11	Total
Girls	14	25	39
Boys	9	?	46
Total	23	62	85

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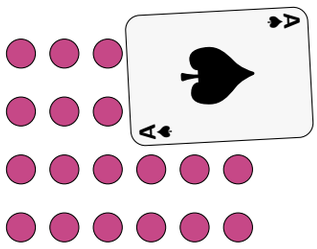
4. 在一个用圆点排成的点阵中，某些圆点被一张牌遮住了。
请问被遮住的圆点占全部的几分之几？

The circles are in a regular rectangular pattern. Some circles are hidden by the card.
What fraction of the circles is hidden?



(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{6}$ (E) $\frac{1}{18}$

---

5. 请问下列哪一个选项内的数值最大？

Which one of the following is the largest number?

(A) 4.05 (B) 4.45 (C) 4.5 (D) 4.045 (E) 4.54

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6. 请问 $\frac{1}{2}$ 的 25% 是多少？

What is 25% of $\frac{1}{2}$?

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) 1 (E) 2

---

7. 我们正驾车从 E 市到 R 市，刚离开时我们看到右侧之标示牌。
我们希望在前往 R 市接近一半路途的地方停下来午餐并休息。
请问在哪一个城市停下来最合适？

(A) G 市 (B) N 市 (C) T 市 (D) B 市 (E) W 市

We're driving from Elizabeth to Renmark, and as we leave we see this sign.
We want to stop at a town for lunch and a break, approximately halfway to Renmark.
Which town is the best place to stop?

(A) Gawler (B) Nuriootpa (C) Truro (D) Blanchetown (E) Waikerie

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8. 将字母 F 先以顺时针方向旋转 90°，然后以水平线为轴做镜像。
请问最后看起来是哪一个选项内的样子？

This letter F is first rotated by 90° clockwise and then reflected in a horizontal line. It will now look like this.

(A) (B) (C) (D) (E)

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9. 小伊在一张纸上写下从 1 到 20 的整数。请问带1的数字她要写几次？

Edith wrote down the whole numbers from 1 to 20 on a piece of paper. How many times did she write the digit 1?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

---

10. 小丹用计算器将一个正整数 P 除以另一个正整数 Q，得到的结果为 3.125。
过一阵子，小丹忘记这两个正整数，只记得它们都小于 30。
请问 Q 的值是什么？

Danny divided a whole number P by another whole number Q on his calculator and got the answer 3.125.
Later, Danny forgot the two whole numbers, but he knew that both were under 30.
The value of Q is

(A) 5 (B) 7 (C) 8 (D) 10 (E) 25

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 在这个 3×3 的方格表的每一行每一列都必须包括数 1、2、3 各一个。
请问 $N + M$ 之值是什么？

1		
	2	N
		M

Every row and every column of this 3×3 square must contain each of the numbers 1, 2 and 3.
What is the value of $N + M$?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

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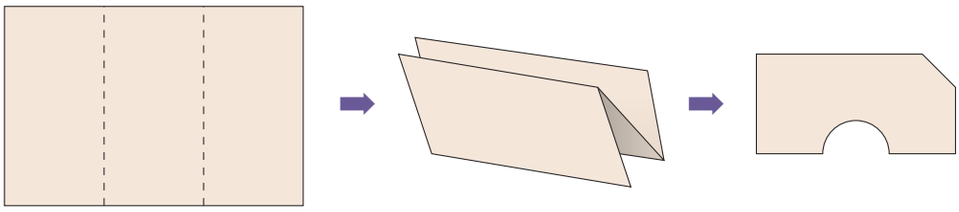
12. 将一张纸折成三等分，然后剪出一个半圆并直着剪一角，如下图所示。

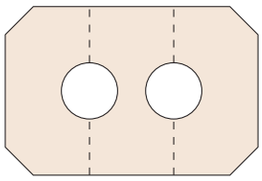


当将这张纸展开，请问它看起来是哪一个选项？

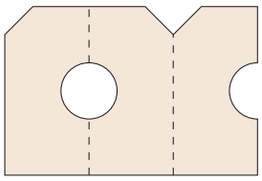
A piece of paper is folded in three, then a semi-circular cut and a straight cut are made, as shown in the diagram.

When the paper is unfolded, what does it look like?

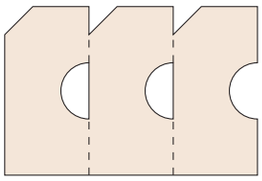
(A)



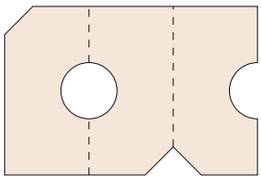
(B)



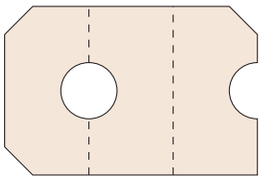
(C)



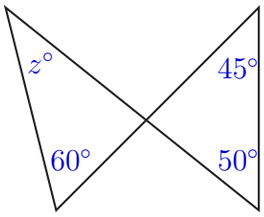
(D)



(E)



13. 请问 z 之值是什么？



What is the value of z?

(A) 30 (B) 35 (C) 45 (D) 50 (E) 55

14. 算式 $\dfrac{111111111}{111} =$

$\dfrac{111111111}{111} =$

(A) 11111 (B) 1001001 (C) 10001 (D) 10101 (E) 1001

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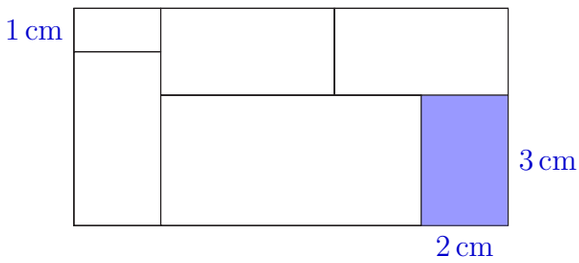
15. 小珍的兄弟数量与姊妹的数量相同。她的哥哥小杰的姊妹数量是兄弟数量的两倍。请问这一家庭总共有多少位小孩？

Jill has the same number of brothers as she has sisters. Her brother Jack has twice as many sisters as he has brothers. How many children are in the family?

(A) 4 (B) 5 (C) 7 (D) 9 (E) 11

---

16. 将一个大矩形分割成 6 个小矩形，如图所示。在右下角涂上阴影的矩形之尺寸为 $2\,\text{cm} \times 3\,\text{cm}$，其它的五个矩形的长都是宽的两倍，而最小的宽等于 $1\,\text{cm}$。



请问原来的大矩形之总面积为多少 cm^2？

The large rectangle shown has been divided into 6 smaller rectangles. The shaded rectangle in the bottom-right corner has dimensions of $2\,\text{cm} \times 3\,\text{cm}$. The remaining five rectangles all have the long side equal to twice the short side. The smallest of these has a width of $1\,\text{cm}$.

What is the total area of the original large rectangle, in square centimetres?

(A) 42 (B) 44 (C) 50 (D) 56 (E) 70

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17. 在我的舞蹈课里，有 14 位学生比小柏高、有 12 位比小艾矮。有四位学生都比小艾矮也都比小柏高。请问我的舞蹈课里共有多少位学生？

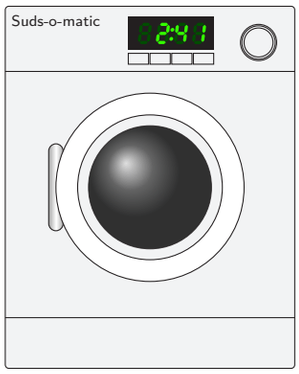
In my dance class, 14 students are taller than Bob, and 12 are shorter than Alice. Four students are both shorter than Alice and taller than Bob. How many students are in my dance class?

(A) 22 (B) 24 (C) 26 (D) 28 (E) 30

18. 我的洗衣机有一个数字显示器，它倒数计时显示截至清洗完毕还剩多少时间，然而有些时候会让我把剩余时间与当前的时刻混淆。

在昨天下午 1：05 的时候，洗衣机显示 2：41，即还剩下 2 小时 41 分。

当洗衣机倒数计时，请问在什么时刻显示器会与当时准确的时刻相吻合？



(A) 下午 2：41 (B) 下午 3：46 (C) 下午 2：23 (D) 下午 1：36 (E) 下午 1：53

My washing machine has a digital display. It counts down the time remaining until the end of the wash, although sometimes I confuse the time remaining with the actual time.

At 1:05 pm yesterday the washing machine displayed 2:41, namely 2 hours and 41 minutes remaining.

When did the washing machine's countdown display happen to agree with the actual time?

(A) 2:41 pm (B) 3:46 pm (C) 2:23 pm (D) 1:36 pm (E) 1:53 pm

19. 有一个形如 $20AMC19$ 的七位数，它的所有的数码都互不相同。若这个七位数可以被 9 整除，请问 $A+M+C$ 之值是什么？

A seven-digit number is in the form $20AMC19$, with all digits different. It is divisible by 9.

What is the value of $A+M+C$?

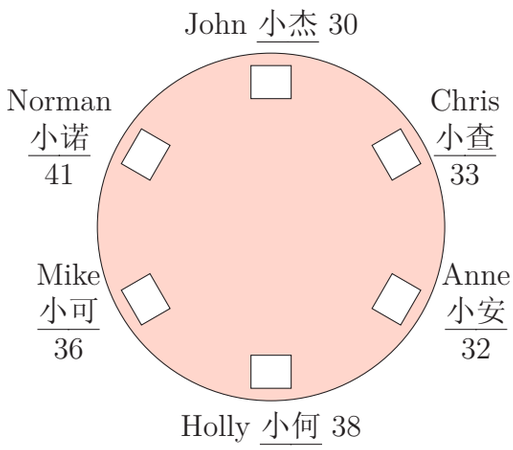
(A) 6 (B) 9 (C) 12 (D) 15 (E) 18

20. 小杰、小查、小安、小何、小可、小诺围坐在一个圆桌前，每个人面前都有一张写有一个数的卡片。

每个人都可以看到他们两位邻居卡片上的数，每个人都喊出两位邻居卡片上的数之和。

已知小杰喊 30、小查喊 33、小安喊 32、小何喊 38、小可喊 36、小诺喊 41。

请问小何面前卡片上的数是什么？



John, Chris, Anne, Holly, Mike and Norman are seated around a round table, each with a card with a number on it in front of them. Each person can see the numbers in front of their two neighbours, and says the sum of these two numbers.

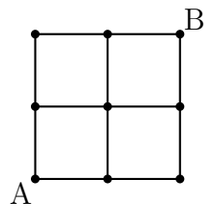
John says 30, Chris says 33, Anne says 32, Holly says 38, Mike says 36 and Norman says 41. What number does Holly have in front of her?

(A) 17 (B) 18 (C) 19 (D) 23 (E) 37

21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 在这个便捷的道路系统中，若不可以通过任何交点超过一次，请问从点 A 到点 B 总共可以有多少条不同的路径？

On this simple system of roads, how many ways are there to get from A to B without visiting any of the 9 intersections more than once?



(A) 8 (B) 10 (C) 12 (D) 14 (E) 16

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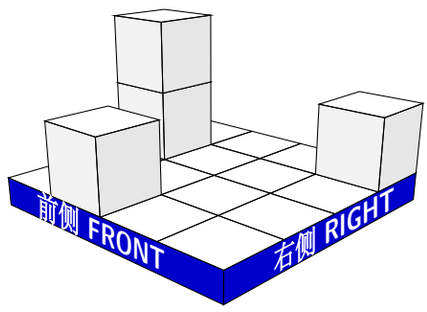
22. 数学课的 30 位学生从家里到学校所花费的平均时间为 21 分钟，而男学生的平均时间为 25 分钟、女学生的平均时间为 19 分钟。请问这个班级里共有多少位男学生？

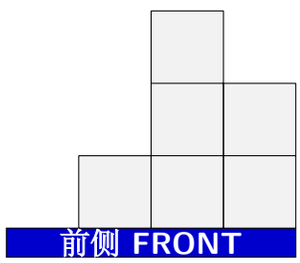
The average time for a class of 30 mathematics students to travel to school is 21 minutes. The boys' average is 25 minutes and the girls' average is 19 minutes. How many boys are in the class?

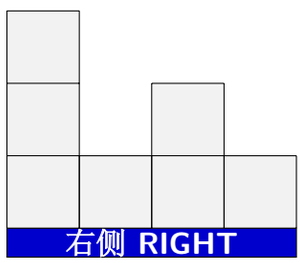
(A) 10 (B) 12 (C) 14 (D) 15 (E) 18

---

23. 在一个 $4\,\text{cm} \times 4\,\text{cm}$ 的平台上放置有一些 $1\,\text{cm}^3$ 的正立方体，如图所示。
现在将这个平台净空，然后放上若干个正立方体，它的前视图与右视图如下图所示。
请问在这个平台上最多可以有多少个正立方体？







A $4\,\text{cm} \times 4\,\text{cm}$ board can have $1\,\text{cm}^3$ cubes placed on it as shown.
The board is cleared, then a number of these cubes are placed on the grid. The front and right side views are shown.
What is the maximum number of cubes there could be on the board?

(A) 10 (B) 11 (C) 16 (D) 17 (E) 18

---

24. 三位运动选手 A、B、C 参加 100 m 赛跑，每人都以匀速跑步。选手 A 最后以 10 秒钟的成绩获胜。

当选手 A 抵达终点时，选手 B 落后他 10 m 。当选手 B 抵达终点时，选手 C 落后他 10 m 。

请问当选手 A 抵达终点时，选手 C 落后他多少 m？

Three athletes Andy, Bob and Chase took part in a 100-metre race, each running at a constant speed. Andy won the race in 10 seconds.

When Andy crossed the finish line, Bob was 10 metres behind. When Bob crossed the finish line, Chase was 10 metres behind Bob.

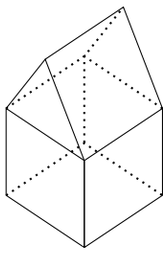
When Andy crossed the finish line, how far behind was Chase?

(A) 21 m (B) 20 m (C) 19 m (D) 18 m (E) 17 m

---

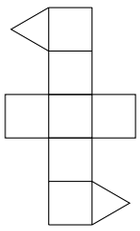
25. 七个正方形与两个正三角形的边长都相等，用它们构造成一个立体的房屋形状，如图所示。

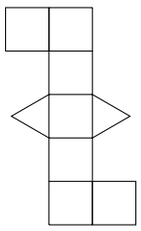
将这个立体沿它的棱边剪开，然后展开在平面上，请问下列哪一个选项不可能是它的展开图？

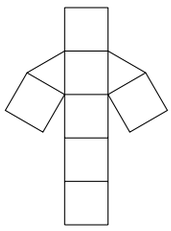


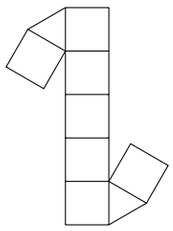
Seven squares and two equilateral triangles, all with the same side lengths, are used to form the 3-dimensional 'house shape' shown.

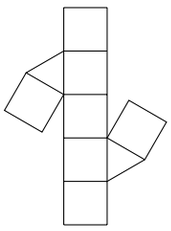
Which of the following diagrams does *not* show a net which can be created by cutting along some of the edges and folding the shape flat?

(A) 

(B) 

(C) 

(D) 

(E) 

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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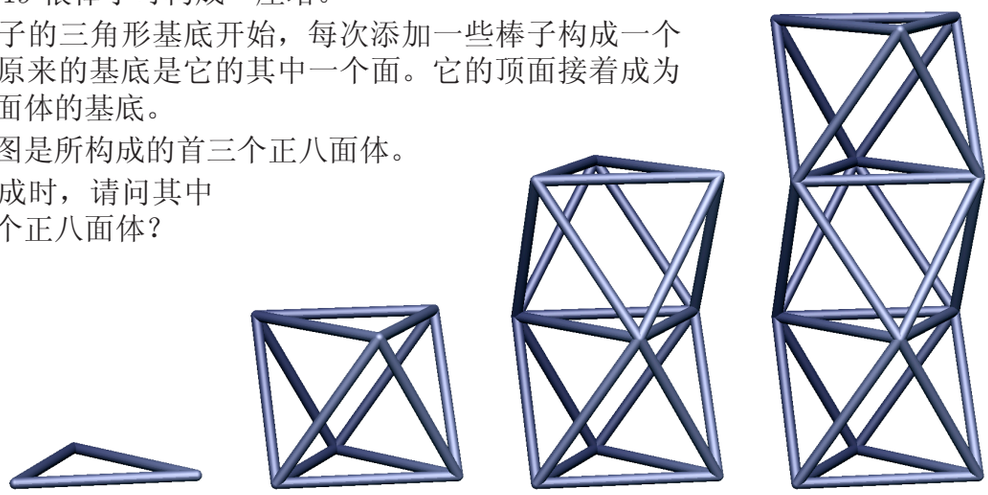
26. 恰好使用 2019 根棒子可构成一座塔。
从用三根棒子的三角形基底开始，每次添加一些棒子构成一个正八面体，原来的基底是它的其中一个面。它的顶面接着成为下一个正八面体的基底。
以下所示的图是所构成的首三个正八面体。
当这座塔完成时，请问其中总共有多少个正八面体？



A tower is built from exactly 2019 equal rods.
Starting with 3 rods as a triangular base, more rods are added to form a regular octahedron with this base as one of its faces. The top face is then the base of the next octahedron.
The diagram shows the construction of the first three octahedra.
How many octahedra are in the tower when it is finished?

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27. 如果一个正整数至少有一个数码与它在这个数中的位数之值相同，则称这个数码为好数。例如 78247 是个好数，因为数码 4 出现在第 4 位。请问在三位数中有多少个好数？

A positive whole number is called *stable* if at least one of its digits has the same value as its position in the number. For example, 78247 is stable because a 4 appears in the 4th position. How many stable 3-digit numbers are there?

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28. 当我们将一个正整数除以 15，其余数会是从 0 到 14 中的一个整数。当我们将一个正整数除以 27，其余数会是从 0 到 26 中的一个整数。
例如，这个正整数是 100，则余数分别是 10 与 19，它们不相等。
请问从 1 到 1000 中，有多少个数除以 15 与除以 27 所得的余数相同？

When I divide an integer by 15, the remainder is an integer from 0 to 14. When I divide an integer by 27, the remainder is an integer from 0 to 26.
For instance, if the integer is 100 then the remainders are 10 and 19, which are different.
How many integers from 1 to 1000 leave the same remainders after division by 15 and after division by 27?

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29. 在一个数列中，如果有三个连续项的和是一个奇数，则称它们为一个奇和三元组。
例如，我们将 1 到 6 的数写成以下的顺序

$$6 \qquad 4 \qquad 2 \qquad 1 \qquad 3 \qquad 5$$

则其中恰有二个奇和三元组：$(4, 2, 1)$ 与 $(1, 3, 5)$。
以某种顺序写下从 1 到 1000，请问最多可以得到多少个奇和三元组？

In a list of numbers, an *odd-sum triple* is a group of three numbers in a row that add to an odd number. For instance, if we write the numbers from 1 to 6 in this order,

$$6 \qquad 4 \qquad 2 \qquad 1 \qquad 3 \qquad 5$$

then there are exactly two odd-sum triples: $(4, 2, 1)$ and $(1, 3, 5)$.
What is the greatest number of odd-sum triples that can be made by writing the numbers from 1 to 1000 in some order?

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30. 有位暴君下令数码 0 不可以用来计算任何数，因为 0 代表没有任何东西。所以在计数中不允许出现任何数码 0。
因此在这个国家的初始几个计数数为 1、2、3、4、5、6、7、8、9、11、12、$\cdots$，其中第十个数是 11。
当您写下初始一千个该国允许的计数数时，请问最后一个数的末三位数码是什么？

The Leader of Zip decrees that the digit 0, since it represents nothing, will no longer be used in any counting number. Only counting numbers without 0 digits are allowed.
So the counting numbers in Zip begin 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, ..., where the tenth counting number is 11.
When you write out the first one thousand allowable counting numbers in Zip, what are the last three digits of the final number?

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